Hiding vegetables to reduce energy density: an effective strategy to increase children's vegetable intake and reduce energy intake

Background: Strategies are needed to increase children's intake of a variety of vegetables, including vegetables that are not well liked. Objective: We investigated whether incorporating puréed vegetables into entrées to reduce the energy density (ED; in kcal/g) affected vegetable and energy intake over 1 d in preschool children. Design: In this crossover study, 3- to 5-y-old children ($n = 40$) were served all meals and snacks 1 d/wk for 3 wk. Across conditions, entrées at breakfast, lunch, dinner, and evening snack were reduced in ED by increasing the proportion of puréed vegetables. The conditions were 100% ED (standard), 85% ED (tripled vegetable content), and 75% ED (quadrupled vegetable content). Entrées were served with unmanipulated side dishes and snacks, and children were instructed to eat as much as they liked. Results: The daily vegetable intake increased significantly by 52 g (50%) in the 85% ED condition and by 73 g (73%) in the 75% ED condition compared with that in the standard condition (both $P < 0.0001$). The consumption of more vegetables in entrées did not affect the consumption of the vegetable side dishes. Children ate similar weights of food across conditions; thus, the daily energy intake decreased by 142 kcal (12%) from the 100% to 75% ED conditions ($P < 0.05$). Children rated their liking of manipulated foods similarly across ED amounts. Conclusion: The incorporation of substantial amounts of puréed vegetables to reduce the ED of foods is an effective strategy to increase the daily vegetable intake and decrease the energy intake in young children. This trial was registered at clinicaltrials.gov as NCT01252433.